

Parkinson's Disease Screening: Complete Resource Guide

Research, Models, Datasets, Implementation Options & Key Resources

A consolidated reference guide for building research-grade AI screening systems using voice, fine-motor, cognitive, gait, and multimodal signals.

Quick Summary

The screening modalities listed in this guide are feasible with existing research models and datasets. Voice screening is among the most mature approaches, while gait and motor analysis provide complementary signals. Reported performance varies substantially by dataset, study design, population, and validation method.

Important: Reported accuracy, AUC, sensitivity, and specificity values below are study-specific research results. They should not be interpreted as guaranteed real-world clinical performance.

1. Voice-Based Parkinson's Screening (Microphone)

Model / Architecture	Reported Performance	Key Features
BiLSTM Voice Model	~97% accuracy; AUC 0.95	Jitter, shimmer, MFCCs, pitch
CNN + LSTM Hybrid	~91% accuracy; F1 0.91	Hybrid deep learning on audio
wav2vec 2.0	AUC 0.9135 (ICEBERG)	Foundation speech model; fine-tunable
Whisper	Fine-tunable for PD research	ASR model and speech representations
SeamlessM4T	Fine-tunable for PD research	Multilingual speech model
Explainable ML + SHAP	98% accuracy; AUC 0.991	Clinical interpretability / feature attribution

Key links: PMC voice study: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12562018/> | ICEBERG / foundation-model study: <https://pubmed.ncbi.nlm.nih.gov/40053624/> | Explainable ML: <https://www.frontiersin.org/journals/aging-neuroscience/articles/10.3389/fnagi.2025.1672971/full>

Datasets

Dataset	Approx. Size / Scope	Description
UCI Parkinson's	195 samples	Classic sustained-vowel /a/ dataset
PC-GITA	Multiple tasks	Spanish Parkinson's voice dataset
ICEBERG	Large-scale	Modern speech foundation-model benchmark
Parkinson's Voice Initiative	10,000+ samples	Telephone-quality, large-scale voice data

Dataset links: UCI: <https://archive.ics.uci.edu/ml/machine-learning-databases/parkinsons/> | Parkinson's Voice Initiative: <https://www.parkinsonsvoice.org/science.php>

Systematic Reviews

Voice analysis in Parkinson's disease — systematic review: <https://www.sciencedirect.com/science/article/pii/S09333365725000442>

Survey of open voice datasets for PD screening: https://link.springer.com/chapter/10.1007/978-3-032-07083-8_3

2. Fine-Motor Screening (Camera / Touch / IMU)

Model / System	Reported Performance	Key Features
PDGes System	>94.5% accuracy	Touch + IMU; 108 volunteers; interpretable
MediaPipe Hand Tracking	Research prototype	Finger tapping and hand-pose estimation
Smartphone Tapping Studies	Good UPDRS correlation	Short tapping clips; non-invasive assessment

Implementation tools: MediaPipe Hand Landmarker — https://ai.google.dev/edge/mediapipe/solutions/vision/hand_landmarker | OpenPose — body/hand pose estimation | Custom touch logging — Android/iOS touch-event APIs.
Research links: PDGes: <https://dl.acm.org/doi/abs/10.1145/3585314> | Smartphone tapping study: <https://pubmed.ncbi.nlm.nih.gov/36155539/>

3. Cognitive Screening (Speech + Motor Timing)

Model / Research	Reported Performance	Key Features
AI Cognitive Digital Assessment	AUC 0.95 (meta-analysis)	Speech + motor timing
Multi-modal CDA Systems	Sensitivity 83%; specificity 92%	Cognitive-motor biomarkers

Systematic review / meta-analysis: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5142802

4. Gait & Balance Analysis (Accelerometer + Gyroscope)

Model / Dataset	Reported Performance	Key Features
mPower Dataset	AUC 0.88–0.89	iPhone IMU; daily activities
DREAM Challenge	Various baselines	Wearable sensor data; competition benchmark
Smartphone Gait Analysis	80–95% accuracy	Step time and stride variability
Passive Sensing (Scientific Reports)	86–93% sensitivity/specificity	Accelerometer + typing; 157 subjects

Resources: mPower / Synapse: <https://www.synapse.org/Synapse:syn5566072> | Smartphone PD detection pilot: <https://pubmed.ncbi.nlm.nih.gov/25819808/> | Scientific Reports passive sensing: <https://www.nature.com/articles/s41598-020-78418-8>

5. ALS Voice & Cough Screening (Microphone)

Research Area	Maturity / Performance	Key Features
ALS Dysarthria Models	Moderate accuracy; research stage	Bulbar speech analysis
Cough Analysis Systems	Research stage	Respiratory muscle weakness signals

Reference review: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12649940/>

Note: ALS voice/cough screening is less mature than Parkinson's voice models and generally has smaller datasets. It remains feasible as a research component.

6. Multi-Modal Integrated Systems

System	Reported Performance	Modalities
Parkinson's Voice Initiative	98.6% detection (lab)	Voice only; large-scale
Multi-Modal MDA + CDA	AUC 0.85–0.95	Motor + cognitive
Deep Learning Passive Sensing	86–93% sensitivity/specificity	Accelerometer + typing

Resources: Parkinson's Voice Initiative: <https://www.parkinsonsvoice.org/science.php> | MDA+CDA: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5142802 | Passive sensing: <https://www.nature.com/articles/s41598-020-78418-8>

7. Implementation Recommendations

1. Start with voice screening — Use the UCI Parkinson's dataset. Begin with a conventional feature pipeline or BiLSTM, then consider fine-tuning wav2vec 2.0. Treat 90%+ accuracy as a study-specific target rather than a guaranteed outcome.
2. Add fine-motor screening — Use MediaPipe for finger-tapping analysis, or build PDGes-style touch + IMU logging.
3. Include gait analysis — Use smartphone IMU sensors and extract step timing, stride variability, acceleration, and related features. Use mPower as a research reference.

4. Combine modalities — Fuse voice, motor, cognitive, and gait signals for a broader screening profile. Multimodal fusion can be early, late, or model-level depending on the available data.

Technical Stack Suggestions

Component	Suggested Stack
Voice	PyTorch / TensorFlow + Librosa + Hugging Face (wav2vec 2.0 / Whisper)
Motor	MediaPipe (Python/JS) or custom touch / IMU logging
Gait	IMU feature extraction + scikit-learn / PyTorch
Deployment	TensorFlow Lite for on-device inference, or Flask / FastAPI backend

8. Additional Resources & Key Papers

- Voice-Based Early Diagnosis of PD (2025): <https://pmc.ncbi.nlm.nih.gov/articles/PMC12562018/>
- Explainable AI for PD Voice Detection: <https://www.nature.com/articles/s41598-025-96575-6>
- Speech biomarkers for PD prediction: <https://www.nature.com/articles/s41531-025-00913-4>
- AI-powered speech screening tool (2025): <https://www.news-medical.net/news/20250724/AI-powered-speech-based-screening-tool-offers-new-hope-for-early-Parkinsons28099s-det>
- Telephone-quality voice screening: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6509044/>

GitHub Repository Search Terms

Search GitHub for: "Parkinson's voice detection deep learning", "wav2vec Parkinson's disease", "PDGes smartphone detection", and "mPower gait analysis".

9. Important Notes & Limitations

- All models and systems described here are research-grade and should not be treated as FDA-cleared clinical diagnostic tools.
- Performance can vary with recording quality, language, accent, device, dataset composition, disease severity, and population.
- A model that performs well on a benchmark dataset may perform differently in real-world screening.
- These systems are best framed as screening or risk-flagging tools that can support, not replace, clinical evaluation.
- Privacy, informed consent, secure storage, bias, explainability, and responsible handling of health-related data should be considered from the beginning of implementation.

10. Practical Conversion / Sharing Note

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